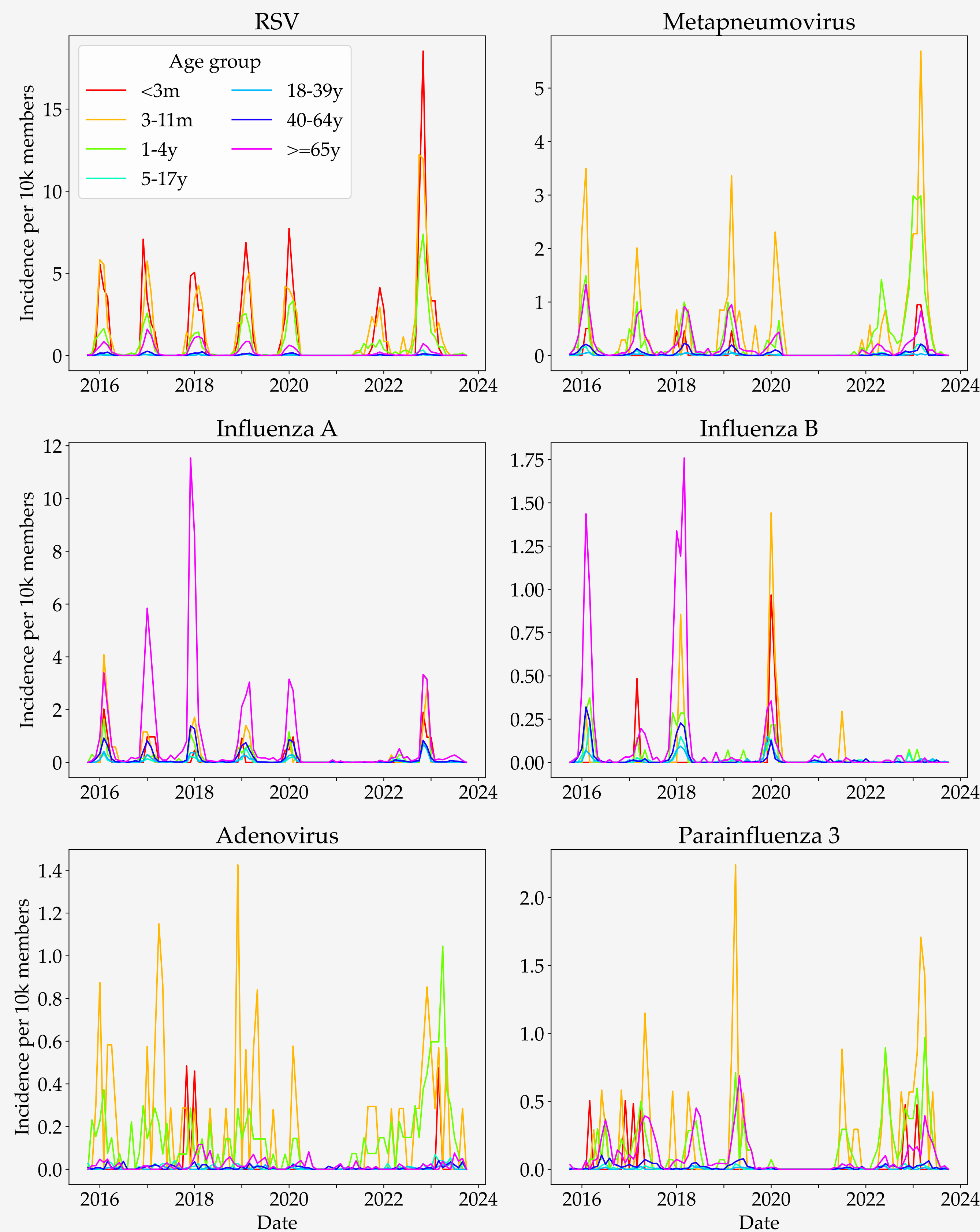
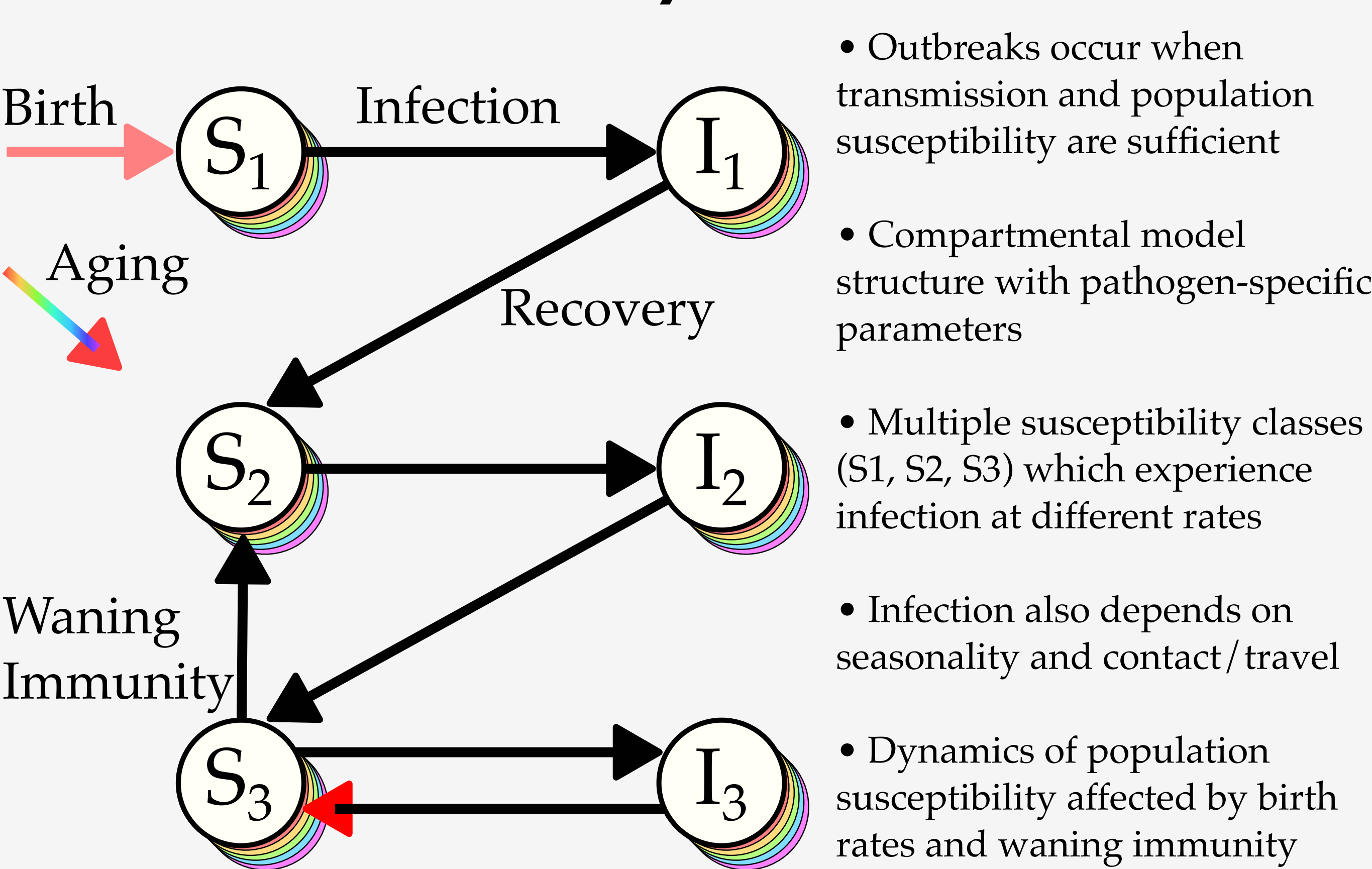


Pathogens rebounded differently

- Kaiser Permanente Southern California data allows us to track respiratory disease incidence using test-confirmed hospitalized cases
- After supression during the pandemic, each pathogen re-emerged differently



Outbreaks have dynamic causes



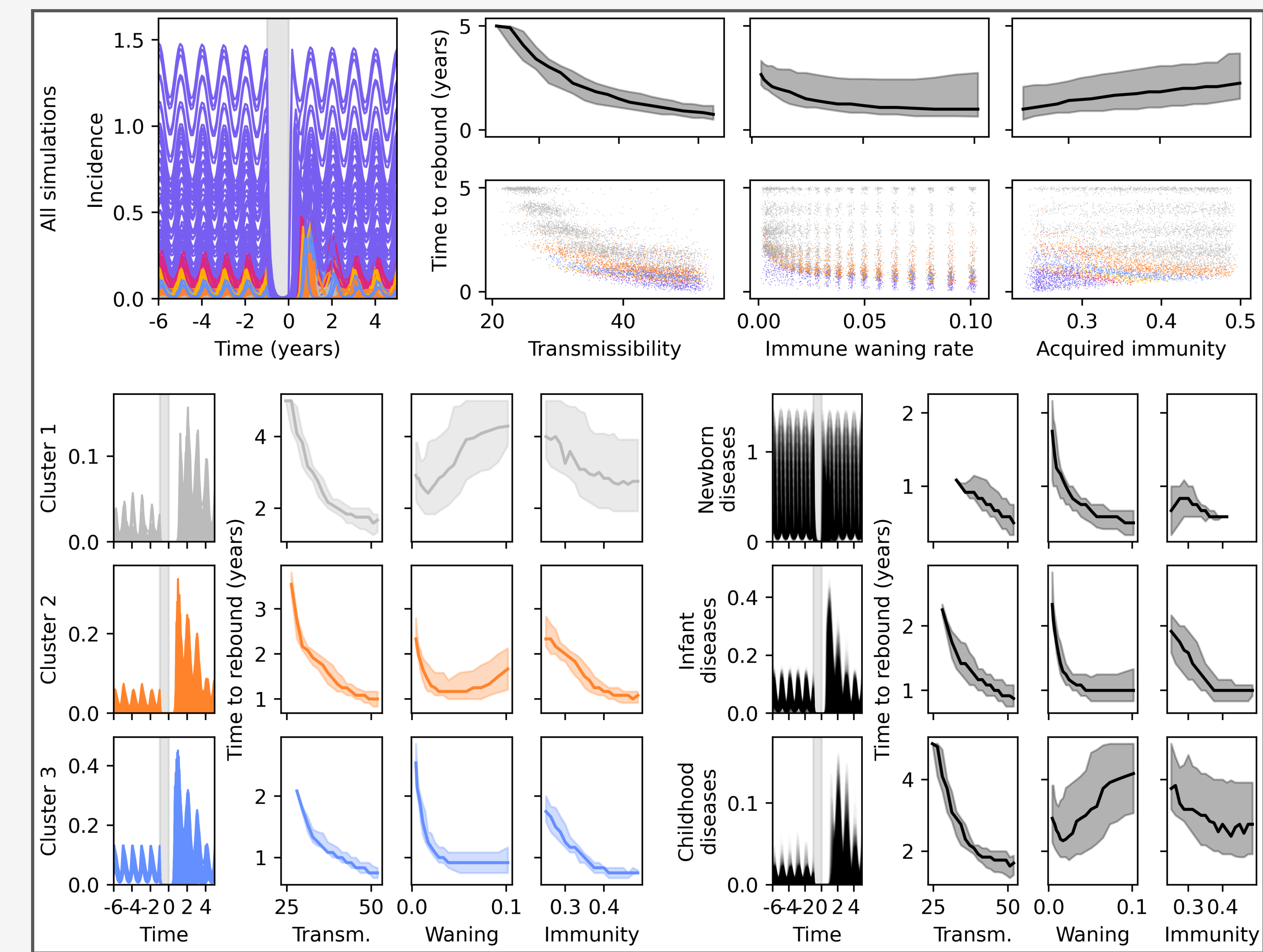
- Outbreaks occur when transmission and population susceptibility are sufficient
- Compartmental model structure with pathogen-specific parameters
- Multiple susceptibility classes (S1, S2, S3) which experience infection at different rates
- Infection also depends on seasonality and contact/travel
- Dynamics of population susceptibility affected by birth rates and waning immunity

A mechanistic model of the post-COVID-19 pandemic re-emergence of respiratory viruses in Southern California

Benjamin J Singer¹, Joseph Lewnard¹, Sara Tartoff²

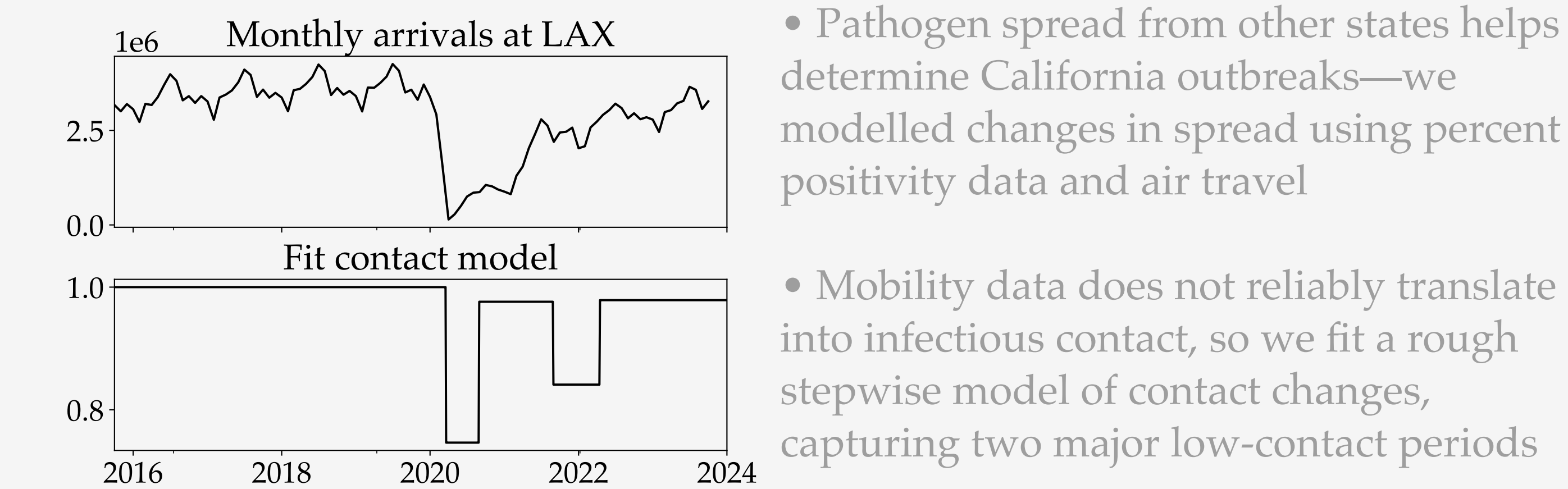
¹ Division of Infectious Diseases, School of Public Health, University of California, Berkeley
² Department of Research and Evaluation, Kaiser Permanente Southern California

Parameters have non-linear effects



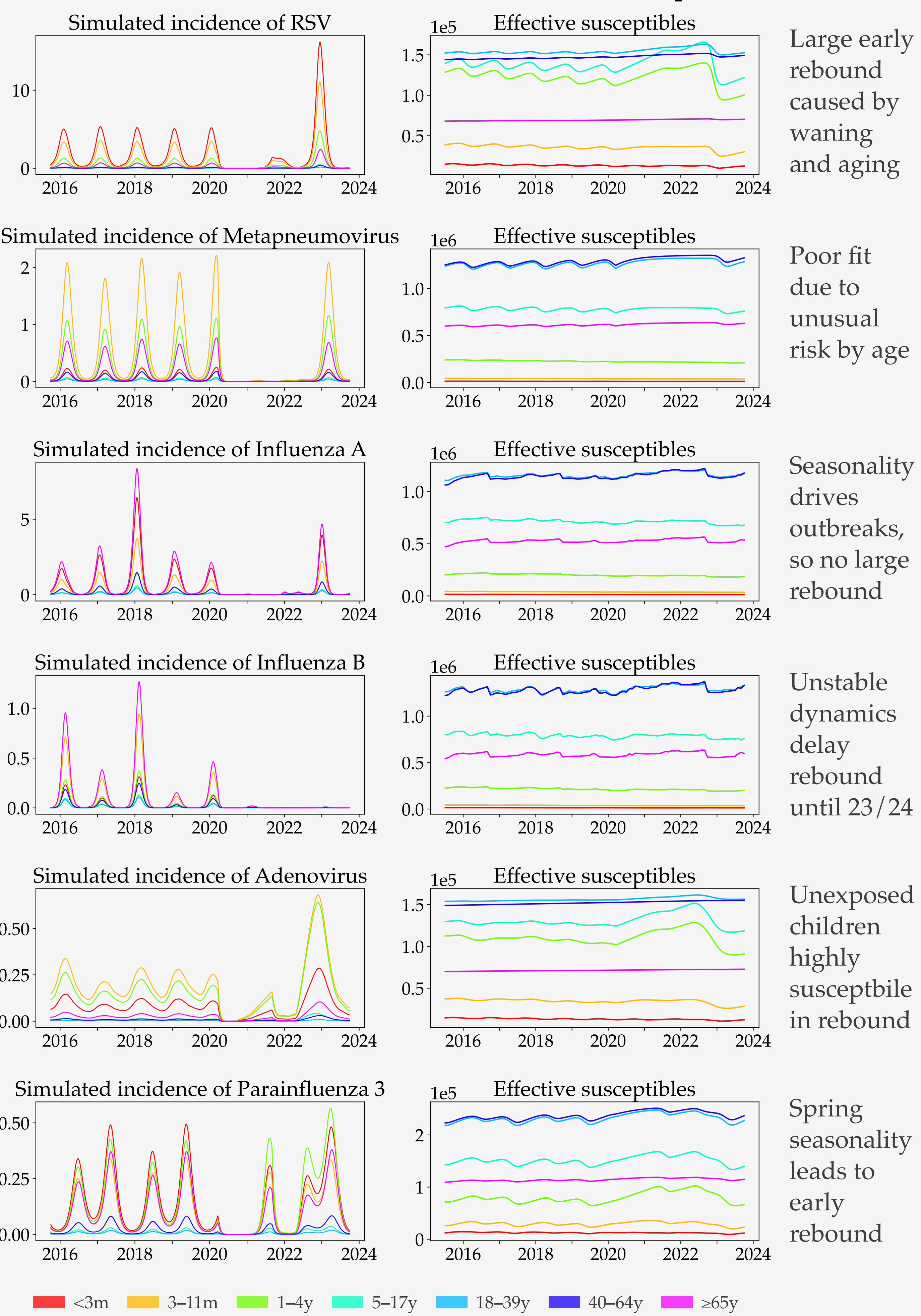
- Multiple parameters can change the time for a simulated disease to rebound after a period of suppression
- Trajectories and points are colored according to a clustering algorithm based on the pre-suppression dynamics
- Acquired immunity has a different effect on time to rebound within these clusters, showing that post-suppression behavior can inform model parameters

COVID-19 reduced contact and travel



- Pathogen spread from other states helps determine California outbreaks—we modelled changes in spread using percent positivity data and air travel
- Mobility data does not reliably translate into infectious contact, so we fit a rough stepwise model of contact changes, capturing two major low-contact periods

Model fits reveal different dynamics



- Large early rebound caused by waning and aging
- Poor fit due to unusual risk by age
- Seasonality drives outbreaks, so no large rebound
- Unstable dynamics delay rebound until 23/24
- Unexposed children highly susceptible in rebound
- Spring seasonality leads to early rebound

Next steps for public health impact

- Models can explain varying rebound behaviours with a mix of population dynamics, waning immunity, and seasonality
- Different diseases respond differently to periods of suppression depending on age strucutre and immunity dynamics
- By quantifying uncertain in the fits we can show what information on disease spread has been gained by measuring rebounds
- In future these models could be deployed in public health to understand disease spread and anticipate disease rebounds after periods of supression